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Revolutionizing Management Information Systems with Natural Language Processing for Digital Transformation

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Abstract

Natural Language Processing (NLP) is a rapidly growing field with immense potential to transform Management Information Systems (MIS) and drive digital transformation in businesses. To explore the possible applications of NLP for MIS, we first investigated current state-of-the-art research by examining the top-tier journals in the field. Our analysis revealed that while there has been some research exploring the benefits of NLP for MIS, the full potential of this powerful technology has not yet been fully realized in the literature. Therefore, this study aims to fill this gap by discussing the potential applications of NLP for MIS. We cover several scenarios, including boosting marketing campaigns, enhancing supplier relationship management, detecting misinformation, using competitive analysis to gain insights, identifying related contextual terms, and enhancing BI systems. The main goal of presenting various scenarios is to demonstrate the vast potential of NLP in transforming MIS and supporting digital transformation. Through this exploration of various scenarios, we hope to inspire researchers and practitioners to further explore the potential of NLP for MIS applications and unlock new possibilities for enhancing business operations, decision-making, and strategy development in the context of digital transformation.

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1. Introduction

Digital transformation is the process of using digital technology to fundamentally change how an organization operates and delivers value to customers [1]. This transformation may involve automating processes, leveraging data analytics, and implementing new digital tools and platforms [1]. Digital transformation has become an increasingly important aspect of modern business operations, as companies seek to adapt to the ever-changing technological landscape. NLP can play a crucial role in this transformation by providing businesses with the tools they need to analyze and leverage the vast amounts of unstructured data that are generated every day [2]. NLP is a subfield of Artificial Intelligence (AI) that focuses on enabling machines to understand, interpret, and generate human language [2]. It involves the use of computational algorithms and techniques to process and analyze large volumes of natural language data (e.g., text).

With NLP, businesses can automate certain processes, such as data analysis for other critical tasks [3]. By integrating NLP into their MIS, businesses can gain a competitive edge by improving their ability to collect, process, and analyze data for decision-making and strategic planning [3]. MIS refers to a set of tools, technologies, and processes used by businesses to optimize their operations, improve performance, and gain a competitive edge in the market [4]. MIS includes various technologies and systems, such as Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) systems, Business Intelligence (BI) systems, Supplier Change Management systems and Knowledge Management systems [4].

Exploring the benefits of NLP for MIS aiming at the digital transformation of business operations is a critical area of research that has gained increasing attention in recent years. There is immense potential for NLP to supplement MIS with valuable data analysis [5]. There are numerous use cases where NLP can provide unique insights into customer preferences, supplier performance, and other critical aspects of business operations. This article highlights six specific scenarios where NLP can enhance MIS. By leveraging the power of NLP, businesses can gain a competitive advantage by unlocking previously untapped sources of data and gaining new insights into their operations and customers.

Section 2 of the paper provides an overview of the study's background, while Section 3 introduces the proposed scenarios. A detailed discussion is presented in Section 4, and the conclusion is provided in Section 5.

2. Background

2.1. Recent research statistics of studies on NLP for MIS

We conducted a targeted search to identify recent developments in NLP for MIS. We focused on research articles published in the last six years in three top-tier MIS journals. Using a set of carefully selected keywords related to NLP including computational linguistics, text analytics, natural language generation, textual analysis, sentiment analysis, speech recognition, Part-of-Speech (POS) tagging, Named Entity Recognition (NER), and Information Retrieval, we searched for relevant keywords in titles and abstracts, and reviewed their abstracts and keywords to ensure their relevance. The number of identified articles related to NLP is presented in Table 1, providing valuable insights for future research in this area.

Table 1. Number of publications related to NLP for MIS.

No.	Journal Name - ISSN	Impact Factor (5Y)	2018	2019	2020	2021	2022	2023
1	Management Information Systems Quarterly - 0276-7783	12.803	2	0	4	0	3	1
2	Information Systems Research - 1047-7047	7.83	2	2	6	2	7	0
3	Journal of the Association for Information Systems - 1536-9323	7.79	1	2	1	2	4	0

Upon examining Table 1, it becomes apparent that there is a lack of research related to NLP in the context of MIS for digital transformation. This may be attributed to the fact that the selected journals primarily focus on research related to MIS systems from a business and management perspective, which may not necessarily encompass the latest developments in NLP. Although there are instances where NLP is being adopted in MIS

systems, the current number of research publications in this area is not substantial, which may indicate a need for further exploration and investigation of the potential of NLP in enhancing digital transformation.

While a significant amount of research has been published on NLP for different information management case studies, which can be found in interdisciplinary journals (e.g., Information Technology and People, Journal of Information Technology, IEEE Transactions on Systems, Man, and Cybernetics: Systems, etc.) that showcase technological advancements. But there is still a gap in the literature when it comes to exploring NLP from a business perspective. Specifically, the literature lacks studies that investigate the potential of NLP in enhancing business processes and boosting overall organizational performance. Addressing this gap can help to enhance the understanding of how NLP can be leveraged to drive digital transformation and improve business outcomes.

2.2. Existing relevant literature

NLP technologies are used to analyse and understand human language [3]. Some key NLP technologies include named entity recognition (NER) (i.e., identifying proper nouns in text), sentiment analysis (i.e., determining the overall emotional tone), topic modelling (i.e., identifying the main themes), and contextual keywords extraction.

NER is a technique in NLP that can be performed using various methods, including gazetteer-based, rule-based, and deep learning-based techniques such as BERT (Bidirectional Encoder Representations from Transformers) [6]. NER has proven useful in a variety of domains. For example, researchers have proposed an NER-based approach to extract key information related to bank accidents, using a BERT-based model [7]. Similarly, NER has been investigated as a means of information extraction from drone flight logs for forensic analysis [8]. Additionally, an NER system has been developed to identify entities mentioned in English business telephone conversations [9].

Sentiment analysis is another area where NLP techniques have been applied. For example, researchers have proposed using sentiment analysis for production prediction based on news [10]. Another study proposed customer review analytics for BI by classifying review sentences as praise or complaint [11]. Yet another study analyzed the emotions and real reactions in Twitter data to gain business insights [12].

When it comes to topic modeling, BERT-based and LDA (Latent Dirichlet Allocation) are both popular approaches. For instance, researchers have used LDA topic modeling to cluster business organizations based on textual data [13]. However, BERT-based [14] is often preferred over LDA because it is a more advanced and powerful language model capable of understanding context and meaning to a greater degree. One study even proposed extracting business insights through dynamic topic modeling and NER [15].

In the field of textual keyword extraction, researchers have explored various techniques [16] for identifying keywords. The most used method is the term frequency–inverse document frequency (TF-IDF) approach [16]. Another popular technique is the TextRank algorithm [17], which has found application in text processing. In recent years, many researchers have combined TF-IDF with TextRank to enhance the effectiveness of keyword extraction [17]. Additionally, an approach called KeyBERT has been proposed [18], which leverages BERT embeddings to generate keywords that are contextually relevant to the document. This minimal and user-friendly technique has several advantages over previous methods, including a greater focus on contextual relevance [18].

3. Showcasing the potential of NLP in MIS through scenario-based presentations

To demonstrate the diverse advantages of NLP in MIS and its potential to support the digital transformation of business operations, we present a comprehensive exploration of six distinct scenarios. These scenarios have been carefully selected after doing an extensive relevant literature review to showcase the varied capabilities of NLP in different aspects of MIS, and to highlight its potential to revolutionize traditional approaches to data processing and analysis. We will study these scenarios as mentioned below.

3.1. Scenario 1: Boosting marketing campaigns through competitor profiling with news data insights.

Imagine a scenario where a company wants to gain an edge over its competitors in the market. By leveraging NLP techniques to extract insights from news data, the company can better understand its rivals' marketing strategies. For example, they could use NLP to extract information about the types of content their competitors are

promoting, the channels they are using to distribute this content, and the target audience they are reaching. By performing competitor profiling with news data insights, the company can identify trends and gaps in the market, allowing them to tailor their marketing campaigns accordingly.

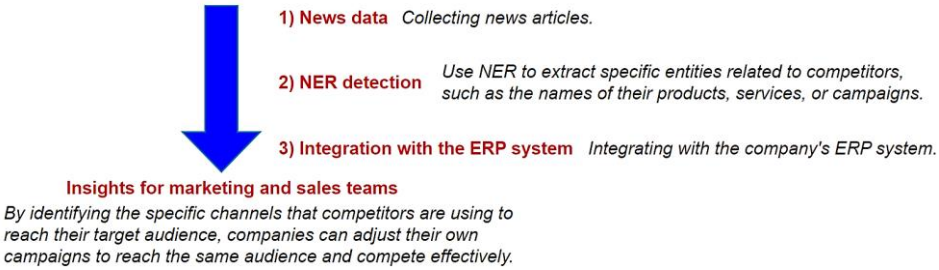

Fig. 1. Using NER for competitor profiling

To conduct competitor profiling for businesses (see Fig. 1), the first step is to gather news data, which can be done using web scraping techniques. Next, NER [19] can be performed to identify key entities such as locations, organizations, and persons (see Fig. 2). By extracting these entities, we can gain insight into the latest contextual information concerning a company. For example, we can see that Puma is partnering with French manufacturers to launch a new collection.

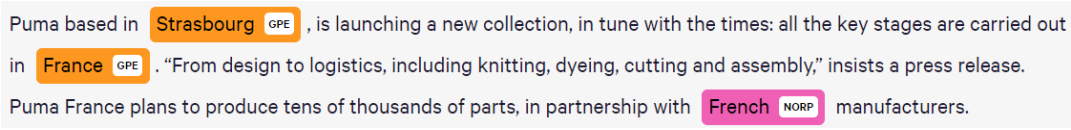

Fig. 2. Identifying named entities using NER

One advantage of using NER is that data analysts do not need to read the entire content of news articles, allowing large collections of articles to be processed efficiently to extract essential information. The results of this analysis can be integrated into the organization's MIS, specifically the ERP system used by marketing and sales teams. This integration provides a comprehensive view of competitors' data, enabling companies to optimize their marketing campaigns and stay ahead of the competition in today's dynamic business landscape.

3.2. Scenario 2: Enhancing supplier relationship management through the collection of updated suppliers' data.

A company manufacturing fire safety equipment that relies on a network of suppliers to provide the raw materials, components, and services required for its operations. However, the procurement department has been struggling to manage the supplier relationships effectively due to the lack of updated information related to the suppliers' business practices, financial stability, and other important factors.

To address this challenge, the company decided to implement a supplier relationship management program that includes the use of news data sources such as industry publications, press releases, and social media posts. By leveraging NLP techniques such as NER and topic detection, the organization can automatically identify and extract relevant information from these sources, including the names of suppliers, their products or services, and any recent news or events related to them (see Fig. 3).

One way to find relevant articles for a company's area of interest is to use topic modeling [5]. This technique can identify key themes within news articles, allowing the latest news to be categorized under the most relevant topics. For example, a news article about FireSafety Inc.'s reputation could be categorized under Topic 4 (see [14] to implement topic modeling) (see Fig. 4). To extract more information, NLP techniques such as NER [19] can be used to validate details such as the company's existence and offered services from the news text (see Fig. 5).

Using this data, the company creates real-time supplier profiles that are accessible to the procurement department and other stakeholders. These profiles provide valuable insights into the suppliers' business practices, financial stability, and any potential risks or opportunities. For example, if a supplier is facing financial difficulties or has been involved in a scandal, the procurement department can quickly identify the potential risks and take appropriate action, such as diversifying its supplier network or renegotiating contracts.

By leveraging these insights, the company can make more informed decisions regarding its supplier relationships, ultimately improving its procurement processes and driving better business outcomes. In addition, the supplier relationship management program can help build stronger partnerships with suppliers by promoting transparency, trust, and collaboration.

Fig. 3. Collecting updated suppliers' data using NLP

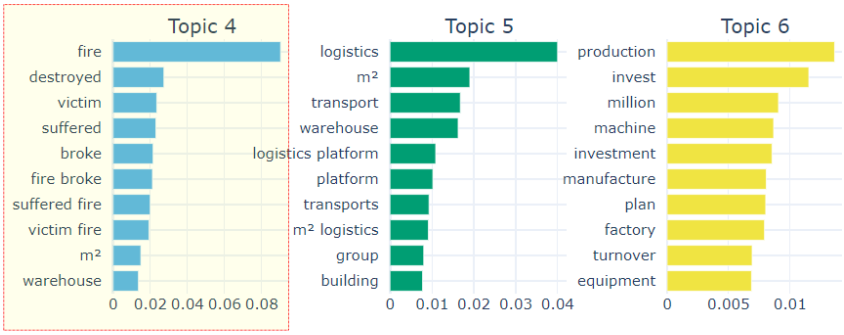

Fig. 4. Identifying the topic of a news article based on the company's existing areas of interest.

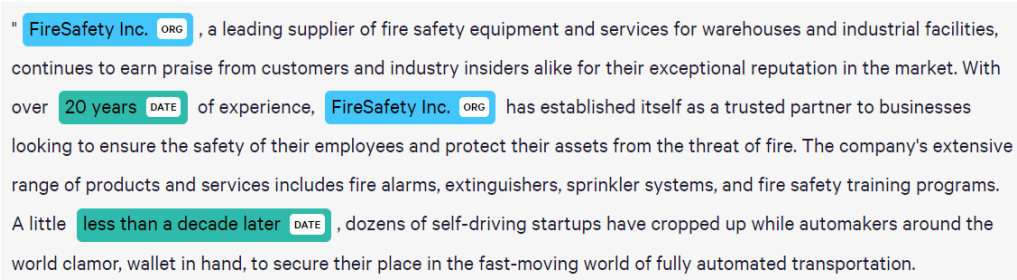

Fig. 5. Identifying named entities using NER

3.3. Scenario 3: Detecting misinformation to prevent reputational harm.

A company that manufactures fire safety equipment (as discussed in Scenario 2) is concerned about potential misinformation being spread about their products. They have been receiving reports of negative news articles and

social media posts that criticize their products and damage their brand reputation. To address this issue, the company decides to leverage NLP techniques (see Fig. 6) to detect and respond to misinformation in news articles.

To start, the company's data science team collects news data from various sources and uses news classification algorithms [20] to categorize the articles as positive (in-favor) or negative (against) towards the organization's products or brand. This step helps the team quickly identify articles that may contain misinformation and require further analysis. Next, the team applies topic detection techniques [14] and NER [19] (see Fig. 4 and Fig. 5) to identify the main topics associated with the organization's products and track any changes that may indicate the spread of misinformation. For instance, if a company's new product is receiving negative feedback in the news, the topic detection algorithms and NER can help identify the specific aspects of the product that are causing the negative sentiment.

Identifying misinformation in news articles is crucial for several teams within the organization. The public relations (PR) and communications teams can use this information to craft messaging that addresses any inaccuracies and clarifies the organization's stance on the issues raised in the news. Marketing teams can also use the insights gained from NLP techniques to adjust their campaigns and messaging to counter any negative perceptions in the market. The legal team can leverage the data collected from NLP techniques to identify and respond to any false claims made in the news that could potentially harm the organization's reputation. Additionally, the executive team can use this data to make data-driven decisions and implement strategies to address any issues that may arise. By detecting and addressing misinformation early on, organizations can protect their brand reputation and make informed decisions to respond appropriately.

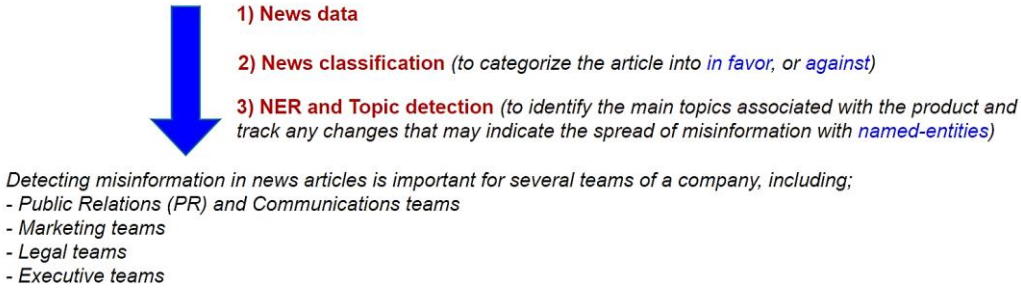

Fig. 6. Identifying misinformation for business

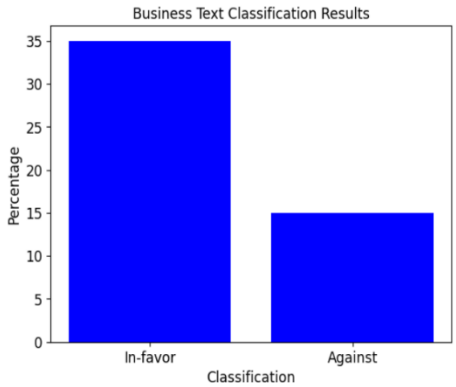

Fig. 6. News text classification

3.4. Scenario 4: Using competitive analysis to gain insights and improve market positioning.

Imagine a business analyst working for a company that sells office equipment. His job is to help the company improve its market position by conducting a competitive analysis. To do this (see Fig. 7), he starts by collecting

online annual reports of the competitors and uses NER [19] (see Fig. 8) to identify key entities such as company names, financial metrics, products, services, and industry terms.

Once he has collected and analyzed the data, he will be able to gain valuable insights into the competitors' performance and market positioning. He discovers that one of your competitors is growing rapidly and has introduced a new line of products that are gaining popularity among customers. Another competitor is struggling to keep up and has experienced a decline in revenue over the past year.

By comparing the company's performance metrics with those of your competitors, he identifies areas of improvement and develops a strategic plan to stay ahead of the competition. To ensure that this information is easily accessible and actionable, he suggest that the company integrates a business dashboard into its MIS, providing a centralized location for competitive analysis, industry trend monitoring, and real-time strategic planning. With these tools at their disposal, the company can make data-driven decisions that lead to continued growth and success, staying ahead of the competition.

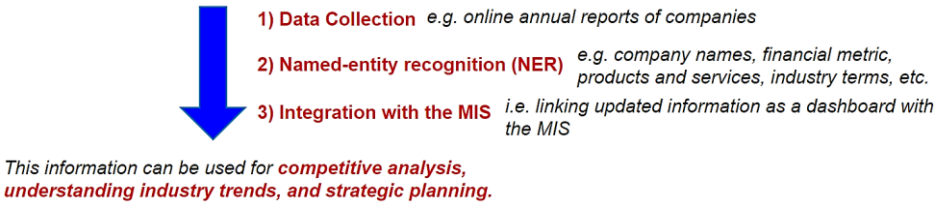

Fig. 7. Implementing competitive analysis on annual reports of companies

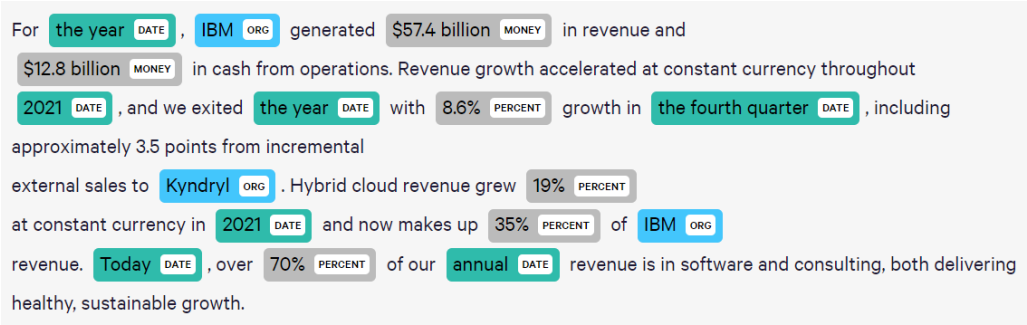

Fig. 8. Using NER to detect relevant entities from a report.

3.5. Scenario 5: Identifying related contextual terms to enrich business taxonomies for MIS Systems.

A large organization that collects vast amounts of data from various sources. However, they are struggling to extract useful insights from their data due to the ever-increasing amount of data being generated. They realize that they need to enrich their business taxonomy with related terms and keywords to create a more comprehensive and accurate understanding of their data by catering to the needs of dynamic business markets.

To achieve this, the organization has decided to leverage NLP methods (see Fig. 9) such as word embedding analysis. For instance, they used Sense2Vec [21] and GloVe [22], which are both popular algorithms used for NLP. Sense2Vec allows you to capture multiple senses of a word [21]. Whereas GloVe stands for Global Vectors for Word Representation and is designed to capture the meaning of words based on their context [22]. These algorithms help to automatically identify and extract key terms from large volumes of text-based data.

By semantically enriching their business taxonomies (see Table 2) with related terms and keywords, the organization gains a better understanding of how their products and services are related to each other [23]. This information can be used to optimize their products and services offerings, improve marketing strategies, and ultimately improve sales.

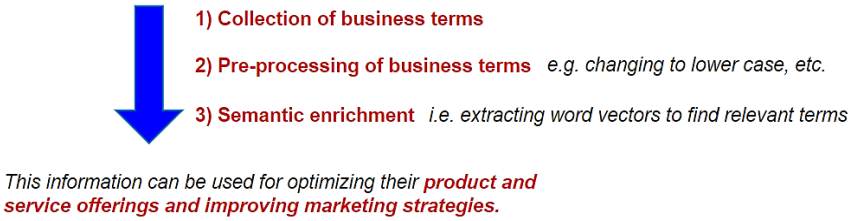

Fig. 9. Finding relevant business terms.

Table 2. Semantic enrichments of business concepts.

No.	Business concept	Semantic enrichment
1	Health insurance	Medical insurance, health coverage, medical coverage
2	Renewable energy	Biofuels, biomass renewables, clean energy, green energy, sustainable energy, fossil fuels
3	Foreign affairs	Foreign relations, foreign policy, international affairs

3.6. Scenario 6: Enhancing BI systems with relevant information retrieval.

This is a large organization that collects vast amounts of data from multiple sources to support their business operations. They are searching for ways to identify similarities and patterns across disparate data. One approach they take is to enrich their MIS systems by identifying the relevance of different texts to each other.

The organization uses NLP techniques to identify shared characteristics between different texts (see Fig. 10). For example, if two texts cover similar context, they may be considered relevant to each other. The organization uses this information to create a more comprehensive understanding of the topic at hand. Similarly, if two texts relate to the same geographic location or occurred within a similar time frame, they may be considered relevant to each other. The organization uses this information to identify patterns that might otherwise go unnoticed. Additionally, if two texts are related to the same organization or written by the same author or group of authors, they may be considered relevant to each other. The organization uses this information to track the performance of different business units, identify potential opportunities for collaboration, or gain deeper insights into the thought processes of key stakeholders.

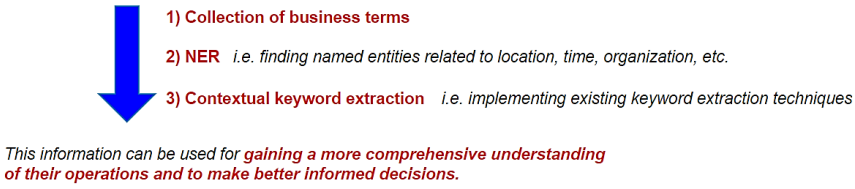

Fig. 10. Finding relevant business terms.

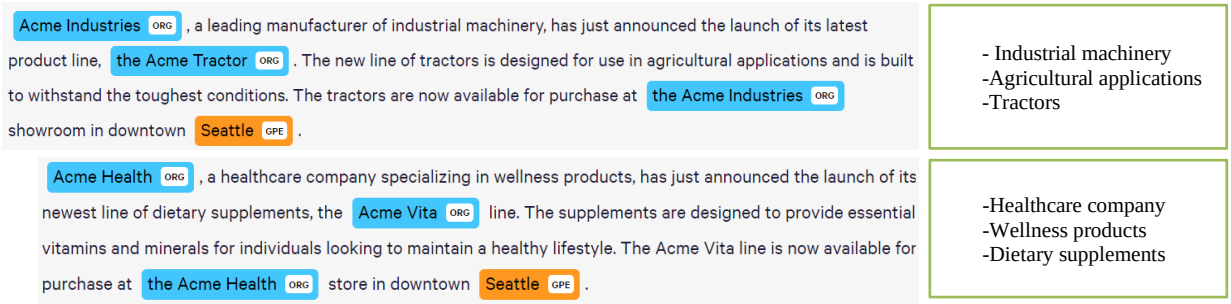

Fig. 11. (a) Identifying named entities (e.g., organizations and locations) [19] in business text (left); (b) Identifying the context using keywords [18] of business texts.

For instance, both texts (in Fig. 11) are about different products from companies with similar names and are in the same city. The first text is about industrial machinery, specifically tractors for agricultural use, while the second text is about dietary supplements for maintaining a healthy lifestyle. By identifying the relevance of different texts to each other, the organization gains a more comprehensive understanding of their operations and can make better-informed decisions.

4. Discussion

With the exponential growth of digital data, businesses are finding it more challenging to extract meaningful insights from the vast amounts of unstructured data available. This is where NLP comes in, allowing businesses to analyze and interpret large volumes of text data in real-time, providing valuable insights that can be used to drive business decisions.

By supplementing MIS with NLP, businesses can gain a deeper understanding of their customers, suppliers, and competitors, which can help them make more informed decisions. For example, by analyzing news data, businesses can gain insights into market trends, enabling them to adjust their marketing campaigns and improve their market positioning. In addition, NLP can help businesses manage their supplier relationships more effectively by analyzing data from multiple sources. This data can provide insights into supplier performance, potential risks, and other factors that can impact a business's supply chain. Moreover, NLP can help businesses prevent reputational harm by identifying and addressing misinformation before it can damage their brand reputation.

Another way NLP can enhance MIS is by identifying related contextual terms that can enrich business taxonomies. By analyzing text data, NLP can identify related terms and concepts, improving the accuracy and relevance of business taxonomies. Finally, by enhancing BI systems with NLP, businesses can gain more accurate insights from unstructured data. This can help them make more informed decisions and improve their overall business performance.

To demonstrate the proof-of-concept demonstration of different business scenarios, the article mentions the use of NER, topic modeling, and word embeddings techniques. While the implementation of NLP techniques is not discussed, an overview is provided to showcase the utility of NLP for MIS in digital transformation. The researchers and practitioners are open to use any technology as long as they serve the same purpose of information extraction.

5. Conclusion

Digital transformation is crucial for businesses to remain competitive in today's fast-paced technological landscape. NLP can be a game-changer in this transformation by providing businesses with powerful tools to analyze and leverage the vast amounts of unstructured data generated every day. By integrating NLP into MIS, businesses can gain a competitive advantage by improving their ability to collect, process, and analyze data for decision-making and strategic planning. NLP can enable MIS to process and analyze unstructured data, leading to better decision-making and more effective strategy development. The potential use cases of NLP for MIS are numerous, including boosting marketing campaigns, enhancing supplier relationship management, detecting misinformation, using competitive analysis to gain insights, identifying related contextual terms, and enhancing BI systems. By leveraging NLP, businesses can unlock previously untapped sources of data and gain new insights into their operations and customers, ultimately leading to improved performance and a better competitive edge in the market.

Acknowledgements

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